Customer Shopping Behavior Analysis

Insights · Segments · Recommendations

3,900 Transactions | 18 Variables | 4 Product Categories

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

Attribute	Detail
Total Records	3,900 rows
Total Columns	18 features
Missing Data	37 values in Review Rating column
Customer Demographics	Age, Gender, Location, Subscription Status
Purchase Details	Item, Category, Amount (USD), Season, Size, Color
Behavioral Features	Discount Applied, Promo Code, Previous Purchases, Frequency, Rating, Shipping

3. Exploratory Data Analysis (Python)

Data preparation and cleaning was performed in Python with the following steps:

- **Data Loading** Imported the dataset using pandas and verified structure with df.info() and .describe().
- **Missing Data Handling** 37 missing values in Review Rating were imputed using the median rating per product category.
- **Column Standardization** Renamed all columns to snake_case for readability and documentation consistency.
- **Feature Engineering** Created age_group (binned ages) and purchase_frequency_days from raw purchase data.
- **Data Consistency Check** Confirmed discount_applied and promo_code_used were redundant; dropped promo_code_used.
- **Database Integration** Connected Python script to SQL Server and loaded the cleaned DataFrame for SQL analysis.

4. Data Analysis using SQL

Structured queries in SQL Server answered the following key business questions:

4.1 Revenue by Gender

Gender	Total Revenue (USD)
Female	$75,191
Male	$157,890

4.2 High-Spending Discount Users

839 customers who used discounts still spent above the average purchase amount — demonstrating that discounts do not suppress premium spending in this segment.

4.3 Top 5 Products by Average Rating

Rank	Item Purchased	Avg. Review Rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4.4 Shipping Type Comparison

Shipping Type	Avg. Purchase Amount (USD)
Standard	$58.46
Express	$60.48

4.5 Subscribers vs. Non-Subscribers

Subscription Status	Total Customers	Avg. Spend (USD)	Total Revenue (USD)
Yes	1,053	$59.49	$62,645
No	2,847	$59.87	$170,436

4.6 Discount-Dependent Products

Item Purchased	Discount Rate (%)
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

4.7 Customer Segmentation

Customer Segment	Number of Customers
Loyal	3,116
Returning	701
New	83

4.8 Top 3 Products per Category

Rank	Category	Item Purchased	Total Orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161

Rank	Category	Item Purchased	Total Orders
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

4.9 Repeat Buyers & Subscriptions

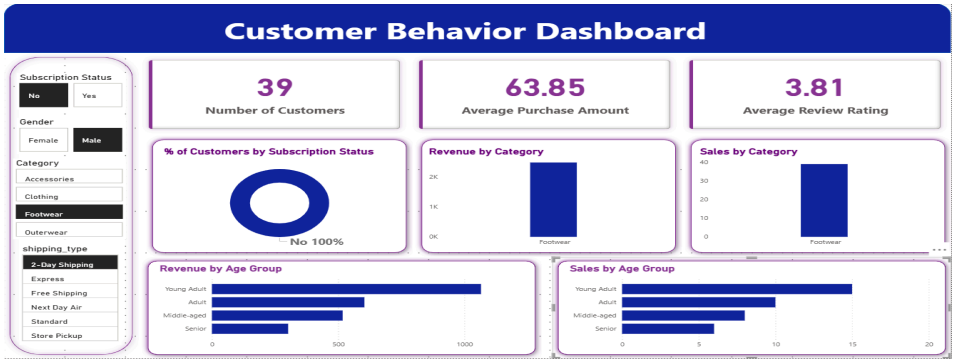
Among customers with more than 5 purchases, 2,518 are non-subscribers vs. 958 subscribers — indicating that purchase frequency alone does not strongly predict subscription uptake.

4.10 Revenue by Age Group

Age Group	Total Revenue (USD)
Young Adult	$62,143
Middle-aged	$59,197
Adult	$55,978
Senior	$55,763

5. Dashboard in Power BI

An interactive dashboard was built in Power BI presenting the following KPIs and visuals:



6. Business Recommendations

Based on the analysis, the following strategic actions are recommended:

1. **Boost Subscriptions –** Subscribers spend nearly as much as non-subscribers ($59.49 vs $59.87). Promote exclusive perks — early access, free shipping, loyalty points — to grow the subscriber base beyond the current 27%.

2. **Customer Loyalty Programs –** With 3,116 customers already classified as Loyal, introduce tiered loyalty rewards to deepen engagement and prevent churn among high-frequency buyers.

3. **Review Discount Policy –** Products like Hat (50%) and Sneakers (49.66%) have near-half discount rates. Audit margins to ensure promotional pricing does not erode profitability while maintaining volume.

4. **Product Positioning –** Gloves, Sandals, and Boots lead on review ratings. Feature these prominently in marketing campaigns and product pages to capitalise on social proof.

5. **Targeted Marketing –** Young Adults ($62,143) are the highest-revenue segment. Tailor digital campaigns and product recommendations to this demographic. Express-shipping users also spend ~$2 more on average — upsell faster delivery at checkout.

6. **High-Value Discount Segment –** 839 discount users still exceeded the average spend. Segment these high-value discount buyers and re-engage them with personalised offers to maximise lifetime value.

7. **Customer Acquisition –** Only 83 New customers vs 701 Returning. Invest in acquisition channels (referral programs, social ads targeting Young Adults) while protecting retention through the existing loyalty pipeline.

8. **Gender Revenue Gap –** Male customers generated $157,890 vs $75,191 for female customers — more than double. Investigate whether this reflects a gender skew in the dataset or a genuine segment opportunity for targeted female-oriented campaigns.